# Work_Sample_Analysis_Clustering

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```
cal <- read.csv("Calendar.csv", header = TRUE)
ft <- read.csv("FileType.csv", header = TRUE)
task <- read.csv("Task.csv", header = TRUE)
uni <- read.csv("Universe.csv", header = TRUE)
```

```
# packages to be preinstalled with the libraries used in this project.

#install.packages("ggplot2")
library(ggplot2)
#library(ggthemes)

#install.packages("dplyr")
library("dplyr")
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library("scales")

library("lubridate")
```

```
##
## Attaching package: 'lubridate'
```

```
## The following object is masked from 'package:base':
##
##     date
```

```
library("stringr")

#install.packages("factoextra")
library("factoextra")
```

```
## Warning: package 'factoextra' was built under R version 3.4.4
```

```
## Welcome! Related Books: `Practical Guide To Cluster Analysis in R` at https://goo.gl/13EFCZ
```

```
summary(cal)
```

```
##        Real         OrgName             Project      
##  Client X:164441   Org A: 2380   Project 91 :  7319  
##                    Org B: 1344   Project 131:  6932  
##                    Org C:51427   Project 96 :  6844  
##                    Org D:18556   Project 137:  6785  
##                    Org E:90734   Project 153:  6203  
##                                  Project 143:  5202  
##                                  (Other)    :125156  
##            Collection        InfraId          eMail       
##  Collection 1118:  2609   Min.   :41051   79120@ :   423  
##  Collection 1478:  1976   1st Qu.:42575   75913@ :   216  
##  Collection 1261:  1894   Median :43336   75759@ :   180  
##  Collection 651 :  1723   Mean   :52253   77201@ :   140  
##  Collection 696 :  1397   3rd Qu.:53746   110091@:   124  
##  Collection 1115:  1296   Max.   :97512   76091@ :   122  
##  (Other)        :153546                   (Other):163236  
##            Developer.ID          Supplier            Country     
##  Developer 79120 :   423   Unknown   :135463   Unknown   :71994  
##  Developer 75913 :   216   Vendor-002: 16287   Country 5 :31035  
##  Developer 75759 :   180   Vendor-004:  3784   Country 17:20522  
##  Developer 77201 :   140   Vendor-003:  2676   Country 16:15013  
##  Developer 110091:   124   Vendor-006:  1360   Country 9 :11555  
##  Developer 76091 :   122   Vendor-042:   621   Country 12: 6724  
##  (Other)         :163236   (Other)   :  4250   (Other)   : 7598  
##       Rank              Segment              Date          Billable.CE   
##  Unknown:73387   Segment 308:70181   05/07/2017:  3803   Min.   :0.000  
```

```
##  G Rank :58569   Segment 595: 1393   07/07/2017:   3801  1st Qu.:0.000  
##  D Rank :20238   Segment 521: 1375   06/07/2017:   3800  Median :0.000  
##  C Rank : 9698   Segment 146: 1136   03/07/2017:   3797  Mean   :1.841  
##  B Rank : 1306   Segment 500:  770   20/07/2017:   3791  3rd Qu.:5.000  
##  E Rank :  685   Segment 469:  658   17/07/2017:   3789  Max.   :5.490  
##  (Other):  558   (Other)    :88928   (Other)   :141660                  
##    Actual.CE      
##  Min.   :    0.000  
##  1st Qu.:    0.000  
##  Median :    0.000  
##  Mean   :    3.968  
##  3rd Qu.:    1.770  
##  Max.   :10962.840  
## 
```

```
summary(cal$Billable.CE)    # it contains numerical values with values between 0 and 5 and is depended on Actual.CE as given,
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   0.000   0.000   0.000   1.841   5.000   5.490
```

```
summary(cal$Billable.CE > 5)  # there are 1745 values greater than 5 which shouldn't be there, let's drop those rows
```

```
##    Mode   FALSE    TRUE 
## logical  162696    1745
```

```
cal <- cal[!cal$Billable.CE > 5,]

sum(cal$Supplier == "Unknown")  # we will remove Unknown at required places
```

```
## [1] 134055
```

```
# Also Developer.ID and eMail are same, so one can be dropped

##

summary(ft)
```

```
##     Enterprise     OrgName             Project     
##  Client X:53826   Org A:  591   Project 91 : 3143  
##                   Org B:  262   Project 96 : 2966  
##                   Org C:16287   Project 137: 2400  
##                   Org D: 8075   Project 144: 1694  
##                   Org E:28611   Project 59 : 1616  
##                                 Project 154: 1394  
##                                 (Other)    :40613  
##           Collection       InfraId           eMail      
##  Collection 1115:  664   Min.   :41081   81062@ :  133  
##  Collection 784 :  562   1st Qu.:42781   79427@ :  126  
##  Collection 1321:  547   Median :43345   135746@:  117  
##  Collection 696 :  543   Mean   :52976   75953@ :  117  
##  Collection 82  :  529   3rd Qu.:63869   75821@ :  109  
##  Collection 160 :  519   Max.   :97512   78921@ :  107  
##  (Other)        :50462                   (Other):53117  
##       Supplier            Country            Rank           FileType    
##  Unknown   :44177   Unknown    :23189   Unknown:23663   java   :13623  
##  Vendor-002: 5314   Country 5 : 9624   G Rank :20098   cs     :10137  
##  Vendor-004: 1121   Country 17: 6513   D Rank : 6284   sql    : 8301  
##  Vendor-003:  872   Country 16: 5330   C Rank : 3051   xml    : 7313  
##  Vendor-006:  594   Country 9 : 3915   B Rank :  333   js     : 6386  
##  Vendor-042:  295   Country 4 : 2008   E Rank :  234   html   : 3393  
##  (Other)   : 1453   (Other)    : 3247   (Other):  163   (Other): 4673  
##          Date          Actual.CE      
##  06/09/2017: 1267   Min.   :  0.000  
##  05/09/2017: 1238   1st Qu.:  0.980  
##  08/08/2017: 1230   Median :  2.610  
##  18/07/2017: 1219   Mean   :  6.142  
##  03/08/2017: 1218   3rd Qu.:  6.500  
##  10/08/2017: 1218   Max.   :943.240  
##  (Other)   :46436
```

```
summary(ft$Actual.CE) # it contains outliers
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   0.000   0.980   2.610   6.142   6.500 943.240
```

```
summary(ft$Actual.CE[!ft$Actual.CE %in% boxplot.stats(ft$Actual.CE)$out])  # without outliers
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   0.000   0.870   2.210   3.424   4.960  14.780
```

```
# eMail can be used instead of Developer.ID over here

##

summary(task)
```

```
##     Enterprise     OrgName            Project
##  Client X:58940   Org A:  725   Project 91 : 3493
##                   Org B:  378   Project 137: 2667
##                   Org C:16742   Project 96 : 2531
##                   Org D: 8143   Project 131: 2122
##                   Org E:32952   Project 153: 2046
##                                 Project 155: 2044
##                                 (Other)    :44037
##            Collection       InfraId          eMail
##  Collection 1118:  834   Min.   :41051   79120@ :  438
##  Collection 1478:  777   1st Qu.:42698   75913@ :  181
##  Collection 1261:  696   Median :43332   75759@ :  148
##  Collection 1115:  603   Mean   :52098   77201@ :  140
##  Collection 651 :  530   3rd Qu.:53746   78921@ :  114
##  Collection 696 :  504   Max.   :97512   77489@ :  102
##  (Other)        :54996                   (Other):57817
##        Supplier           Country           Rank               Date
##  Unknown   :48781   Unknown   :25301   Unknown:25818   06/09/2017: 1385
##  Vendor-002: 5446   Country 5 :10394   G Rank :21280   05/09/2017: 1365
##  Vendor-004: 1327   Country 17: 7105   D Rank : 7210   10/08/2017: 1323
##  Vendor-003:  887   Country 16: 6254   C Rank : 3700   25/07/2017: 1320
##  Vendor-006:  649   Country 9 : 4157   B Rank :  499   03/08/2017: 1314
##  Vendor-042:  288   Country 4 : 2364   E Rank :  224   13/07/2017: 1313
##  (Other)   : 1562   (Other)   : 3365   (Other):  209   (Other)   :50920
##      Task.Id            Task.Type       Actual.CE
##  null    :51812   null       :51812   Min.   :    0.00
##  Task 438:  211   New Feature: 2769   1st Qu.:    1.34
##  Task 49 :  198   Task       : 2746   Median :    4.41
##  Task 290:  179   Improvement:  624   Mean   :   11.07
##  Task 605:  143   Bug        :  460   3rd Qu.:   11.86
##  Task 552:  116   Sub-task   :  283   Max.   :10962.84
##  (Other) : 6281   (Other)    :  246
```

```
##

summary(uni)
```

```
##     Enterprise    OrgName           Project              Collection
##  Client X:7735   Org A: 111   Project 111: 472   Collection 1   :   2
##                  Org B: 359   Project 91 : 371   Collection 100 :   2
##                  Org C:2760   Project 165: 262   Collection 1003:   2
##                  Org D: 798   Project 130: 254   Collection 1008:   2
##                  Org E:3707   Project 137: 243   Collection 101 :   2
##                               Project 154: 209   Collection 1010:   2
##                               (Other)    :5924   (Other)        :7723
##     InfraId          Repository          Type
##  Min.   :41051   Repo 1   :   1   Jira      :2056
##  1st Qu.:43344   Repo 10  :   1   Subversion:5679
##  Median :45613   Repo 100 :   1
##  Mean   :54359   Repo 1000:   1
##  3rd Qu.:54744   Repo 1001:   1
##  Max.   :97513   Repo 1002:   1
##                  (Other)  :7729
```

---

# a # Number of Vendors used in each country

```
str(cal$Supplier)       # it has 45 known vendors and the rest suppliers fall under unknown
```

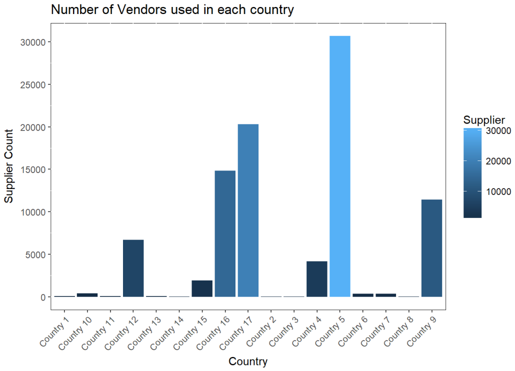
```
##  Factor w/ 46 levels "Unknown","Vendor-002",..: 1 1 1 1 1 1 1 1 1 1 ...
```

```
summary(cal$Supplier == 'Unknown')  # there are 134055 suppliers which do not fall in given vendors
```

```
##    Mode   FALSE    TRUE
## logical   28641  134055
```

```
# as we don't need unknown values over here, let's drop them

a <- aggregate(Supplier ~ Country, data = cal, length)

a_2 <- a[!a$Country == "Unknown", ] # this drops all the rows containing Unknown supplier
```

```
ggplot(data = a_2, aes(Country, Supplier, fill = Supplier)) + geom_bar(stat = "identity") +
        xlab("Country") + ylab("Supplier Count") +
        ggtitle("Number of Vendors used in each country") +
        theme(panel.background = element_rect(fill = "white",
        color = "black", size = 0.1),
        axis.text.x = element_text(angle = 45, hjust = 1)) +
        scale_y_continuous(breaks = pretty_breaks(n = 10))
```



---

# b # the average number of Developers in all the repositories and What is the average number of Repositories in all the Projects

```
# Each repository has a unique Infra ID, so we take InfraId for our calculation as Calendar dataset contains both

# let's look at the structure of InfraId, Developer.ID and Project
str(cal$InfraId)        # it is an integer value,let's see number of unique values
```

```
##  int [1:162696] 83707 83707 83707 83707 83707 83707 83707 83707 83707 83707 ...
```

```
length(unique(cal$InfraId)) # it contains 1589 unique values
```

```
## [1] 1589
```

```
str(cal$Developer.ID)    # it contains 4212 different developers
```

```
##  Factor w/ 4212 levels "Developer 107491",..: 1524 1524 1524 1524 1524 1524 1524 1524 1524 1524 ...
```

```
str(cal$Project)        # it contains 171 projects
```

```
##  Factor w/ 171 levels "Project 1","Project 10",..: 101 101 101 101 101 101 101 101 101 101 ...
```

```
# the average number of Developers in all the repositories

repo_dev <- aggregate(Developer.ID ~ InfraId, data = cal, length)

(sum(repo_dev$Developer.ID)/ length(repo_dev$InfraId))
```

```
## [1] 102.3889
```

```
# the average number of Repositories in all the Projects

proj_repo <- aggregate(InfraId ~ Project, data = cal, length)

(sum(proj_repo$InfraId)/ length(proj_repo$Project))
```

```
## [1] 951.4386
```

# c # the Top 3 Suppliers in terms of Billable Coding Effort (CE) Delivered per developer

```r
# let's look at the structure of the relevant columns

str(cal$Developer.ID)   # it contains 4212 different developers
```

```
##  Factor w/ 4212 levels "Developer 107491",..: 1524 1524 1524 1524 1524 1524 1524 1524 1524 1524 ...
```

```r
str(cal$Billable.CE)    # it contains numerical values with values between 0 and 5 and is depended on Actual.CE as given,
```

```
##  num [1:162696] 5 5 5 5 5 5 5 5 5 5 ...
```

```r
str(cal$Supplier)       # it has 45 known vendors and the rest suppliers fall under unknown
```

```
##  Factor w/ 46 levels "Unknown","Vendor-002",..: 1 1 1 1 1 1 1 1 1 1 ...
```

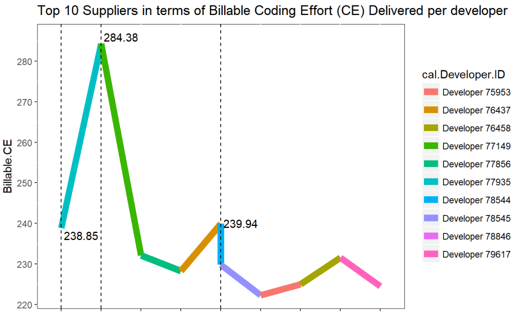
```r
summary(cal$Supplier == 'Unknown')  # there are 134055 suppliers which do not fall in given vendors
```

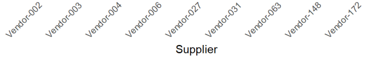
```
##    Mode   FALSE    TRUE
## logical   28641  134055
```

```r
# as we don't need unknown values over here, let's drop them

c <- data.frame(cal$Developer.ID, cal$Billable.CE, cal$Supplier)

c_2 <- c[!c$cal.Supplier == "Unknown", ]  # this drops all the rows containing Unknown supplier
```

```r
sup_bill <- aggregate(cal.Billable.CE ~ cal.Supplier + cal.Developer.ID, data = c_2, sum)

# let's arrange sup_bill according to Billabe_CE

# library(dplyr)
sup_bill_10 <- head(arrange(sup_bill, desc(cal.Billable.CE)), n = 10)  # We get top 6 suppliers according to their Billable.CE

# Supplier       Developer.ID      Billable.CE
# Vendor-003      Developer 77149       284.38
# Vendor-027    Developer 78544      239.94
# Vendor-002      Developer 77935       238.85
```

```r
# library(scales)

ggplot(sup_bill_10,aes(x = cal.Supplier, y = cal.Billable.CE, fill = cal.Developer.ID, group = 1)) +
  geom_line(stat = "identity", aes(colour = cal.Developer.ID), size = 3) +
  scale_y_continuous(breaks = pretty_breaks(n = 10)) +
  labs(title = "Top 10 Suppliers in terms of Billable Coding Effort (CE) Delivered per developer") +
  xlab("Supplier") + ylab("Billable.CE") +
  geom_vline(xintercept = 1, linetype = "dashed") +
  annotate("text", label="238.85", colour="black", x = 1.5, y = 237) +
  geom_vline(xintercept = 2, linetype = "dashed") +
  annotate("text", label="284.38", colour="black", x = 2.5, y = 286) +
  geom_vline(xintercept = 5, linetype = "dashed") +
  annotate("text", label="239.94", colour="black", x = 5.5, y = 240) +
  theme(panel.background = element_rect(fill = "white",
        color = "black", size = 0.2),
        axis.text.x = element_text(angle = 45, hjust = 1))
```





---

# d # the Project with a maximum number of Billable CE hours per day delivered per developer

```
# let's look at the structure of the relevant columns

str(cal$Project)  # it contains 171 different Projects
```

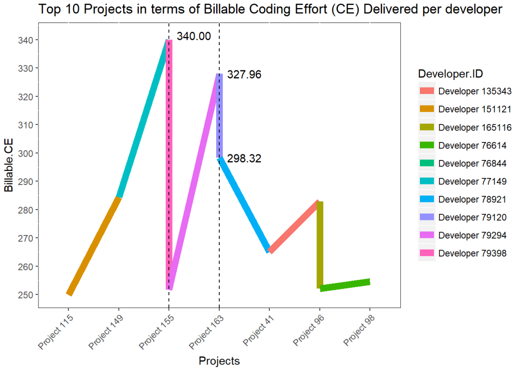
```
##  Factor w/ 171 levels "Project 1","Project 10",..: 101 101 101 101 101 101 101 101 101 101 ...
```

```
proj_bill <- aggregate(Billable.CE ~ Project + Developer.ID, data = cal, sum)

max(proj_bill$Billable.CE)
```

```
## [1] 340
```

```
#    Project      Developer.ID      Billable.CE
#  Project 155  Developer 79398      340.00

proj_bill_10 <- head(arrange(proj_bill, desc(Billable.CE)), n = 10)
```

```
library(scales)

ggplot(proj_bill_10,aes(x = Project, y = Billable.CE, fill = Developer.ID, group = 1)) +
  geom_line(stat = "identity", aes(colour = Developer.ID), size = 3) +
  scale_y_continuous(breaks = pretty_breaks(n = 10)) +
  labs(title = "Top 10 Projects in terms of Billable Coding Effort (CE) Delivered per developer") +
  xlab("Projects") + ylab("Billable.CE") +
  geom_vline(xintercept = 3, linetype = "dashed") +
  annotate("text", label="340.00", colour="black", x = 3.5, y = 341.5) +
  geom_vline(xintercept = 4, linetype = "dashed") +
  annotate("text", label="327.96", colour="black", x = 4.5, y = 327.96) +
  geom_vline(xintercept = 4, linetype = "dashed") +
  annotate("text", label="298.32", colour="black", x = 4.5, y = 298.32) +
  theme(panel.background = element_rect(fill = "white",
        color = "black", size = 0.2),
        axis.text.x = element_text(angle = 45, hjust = 1))
```



---

# e # the difference on the total Actual CE delivered between July and August in Country 17

```
# let's look at the values

str(ft$Actual.CE)
```

```
##  num [1:53826] 2.73 7.61 8.19 0 1.84 0.5 0.94 1.81 5.56 6.62 ...
```

```
# it contains outliers which should be removed

str(ft$Actual.CE[!ft$Actual.CE %in% boxplot.stats(ft$Actual.CE)$out])
```

```
##  num [1:48827] 2.73 7.61 8.19 0 1.84 0.5 0.94 1.81 5.56 6.62 ...
```

```
# summary without outliers
```

```
country_17 <- data.frame(ft$Actual.CE[ft$Country == 'Country 17'], as.factor(month(dmy(ft$Date)))[ft$Country == 'Country 17'
], ft$Country[ft$Country == 'Country 17'])
```

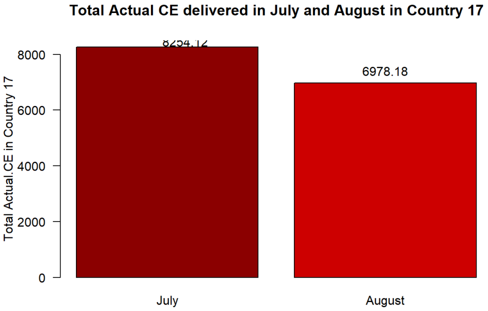
```
# let's remove outliers in Actual.CE
country_17 <- country_17[!country_17$ft.Actual.CE.ft.Country.....Country.17.. %in% boxplot.stats(country_17$ft.Actual.CE.ft.
Country.....Country.17..)$out,]
```

```
country_17_7 <- sum(country_17$ft.Actual.CE.ft.Country.....Country.17..[country_17$as.factor.month.dmy.ft.Date....ft.Countr
y.....Country.17.. == 7])
country_17_8 <- sum(country_17$ft.Actual.CE.ft.Country.....Country.17..[country_17$as.factor.month.dmy.ft.Date....ft.Countr
y.....Country.17.. == 8])
```

```
country_17_diff <- (country_17_8 - country_17_7)
country_17_diff     # difference in total Actual CE between July and August in Country 17 is 1275.94
```

```
## [1] 1275.94
```

```
diff <- c(country_17_8, country_17_7)

barplot(diff,
  main = "Total Actual CE delivered in July and August in Country 17",
  ylim = c(0,8500),
  #horiz = TRUE,
  las = 1,
  names.arg = c("July", "August"),
  col = c("red4","red3"),
  ylab = "Total Actual.CE in Country 17")

text(0.8, 8457, "8254.12")
text(1.9, 7400, "6978.18")
```



## f # the total number of developers in each one of the last 3 months for Vendor-002

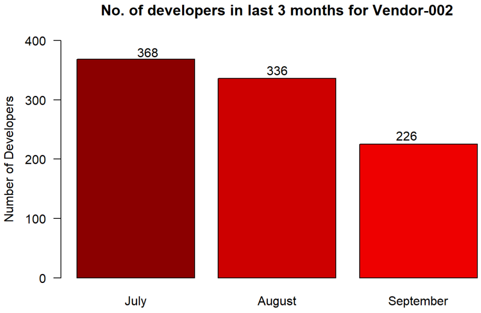
```
# library("lubridate")
f <- data.frame(cal$Developer.ID, cal$Date, cal$Supplier)
f$only_month <- as.factor(month(dmy(f$cal.Date))) # convert month values to factor from numeric
# OR  (above step can also be done like below)
#f$only_month <- month(as.POSIXlt(f$cal.Date, format = "%d/%m/%Y"))
#f$only_month <- as.factor(f$only_month)

str(f)  # we can see the new structure of Date in only_month
```

```
## 'data.frame':    162696 obs. of  4 variables:
##  $ cal.Developer.ID: Factor w/ 4212 levels "Developer 107491",..: 1524 1524 1524 1524 1524 1524 1524 1524 1524 1524 ...
##  $ cal.Date        : Factor w/ 68 levels "01/07/2017","01/08/2017",..: 27 29 31 33 39 41 43 45 47 53 ...
##  $ cal.Supplier    : Factor w/ 46 levels "Unknown","Vendor-002",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ only_month      : Factor w/ 3 levels "7","8","9": 1 1 1 1 1 1 1 1 1 1 ...
```

```
ven_002 <- f[f$cal.Supplier == "Vendor-002",]   # we only need Vendor-002 supplier here, so extract that
```

```
# total number of developers in each one of the last 3 months for Vendor-002
total_dev_7 <- length(unique(ven_002$cal.Developer.ID[ven_002$only_month == 7]))  # 368
total_dev_8 <- length(unique(ven_002$cal.Developer.ID[ven_002$only_month == 8]))  # 336
total_dev_9 <- length(unique(ven_002$cal.Developer.ID[ven_002$only_month == 9]))  # 226
```

```
total_dev <- c(total_dev_7, total_dev_8, total_dev_9)

barplot(total_dev,
  main = "No. of developers in last 3 months for Vendor-002",
  ylim = c(0,400),
  #horiz = TRUE,
  las = 1,
  names.arg = c("July", "August", "September"),
  col = c("red4","red3","red2"),
  ylab = "Number of Developers")

text(0.8, 380, "368")
text(1.9, 350, "336")
text(3.0, 240, "226")
```



---

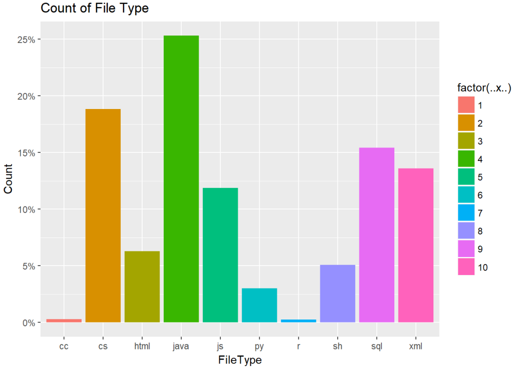
# g # Percentage of the total projects working with Java Files

```
project_java <- ft$Project[ft$FileType == 'java']
prop <- length(project_java)/length(ft$Project)
prop*100    # 25.3 % of total projects worked with Java Files
```

```
## [1] 25.30933
```

```
ggplot(data = ft, aes(FileType, group = length(FileType))) + geom_histogram(aes(y = ..prop.., fill = factor(..x..)), stat="c
ount") +
  labs(title = "Count of File Type") +
  labs(x = "FileType", y = "Count") +
  scale_y_continuous(labels=scales::percent)
```

```
## Warning: Ignoring unknown parameters: binwidth, bins, pad
```



# h # the most common task type in Repository 642

```
# let's look at the structure of the relevant columns

str(uni$Repository)  # it contains 7735 different Repositories
```

```
##  Factor w/ 7735 levels "Repo 1","Repo 10",..: 3143 3295 4900 5416 2219 2390 2392 2498 2605 316 ...
```

```
uni$InfraId[uni$Repository == 'Repo 642'] # 41583 InfraId belongs to Repo 642
```

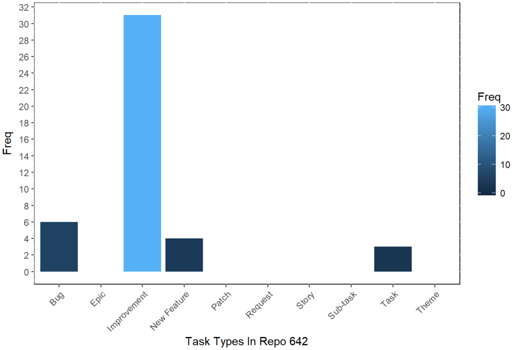
```
## [1] 41583
```

```
task_642 <- task$Task.Type[task$InfraId == 41583] # this stores all the Task Types for relevant InfraId

task_642 <- data.frame(table(task_642)) # this gives a task type null with 107 frequency, which should be ignored

task_642 <- task_642[!task_642$task_642 == 'null',] # dropping null

task_642    # most common task is Improvement with frequency of 31
```

```
##       task_642 Freq
## 1          Bug    6
## 2         Epic    0
## 3  Improvement   31
## 4  New Feature    4
## 6        Patch    0
## 7      Request    0
## 8        Story    0
## 9     Sub-task    0
## 10        Task    3
## 11       Theme    0
```

```
ggplot(data = task_642, aes(x = task_642, y = Freq, fill = Freq)) + geom_bar(stat="identity") +
  scale_y_continuous(breaks = pretty_breaks(n = 15)) +
  xlab("Task Types In Repo 642") +
  ggtitle("Most common task type in Repository 642") +
  theme(panel.background = element_rect(fill = "white",
        color = "black", size = 0.5),
        axis.text.x = element_text(angle = 45, hjust = 1))
```

Most common task type in Repository 642



# Clustering

```r
# store eMail as Developer, as they are same
ft$Developer <- as.factor(gsub("\\@", "", ft$eMail))
```

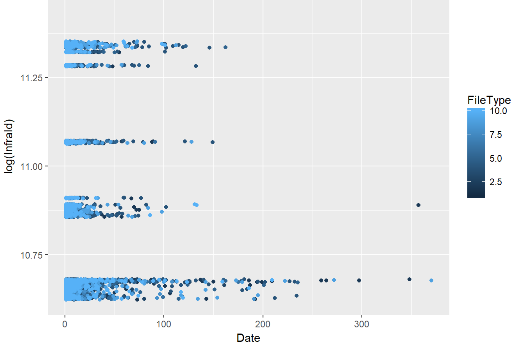
```r
summary(ft)
```

```
##     Enterprise     OrgName            Project     
##  Client X:53826   Org A:  591   Project 91 : 3143  
##                   Org B:  262   Project 96 : 2966  
##                   Org C:16287   Project 137: 2400  
##                   Org D: 8075   Project 144: 1694  
##                   Org E:28611   Project 59 : 1616  
##                                 Project 154: 1394  
##                                 (Other)    :40613  
##            Collection        InfraId           eMail      
##  Collection 1115:  664   Min.   :41081   81062@ :  133  
##  Collection 784 :  562   1st Qu.:42781   79427@ :  126  
##  Collection 1321:  547   Median :43345   135746@:  117  
##  Collection 696 :  543   Mean   :52976   75953@ :  117  
##  Collection 82  :  529   3rd Qu.:63869   75821@ :  109  
##  Collection 160 :  519   Max.   :97512   78921@ :  107  
##  (Other)        :50462                   (Other):53117  
##        Supplier           Country            Rank          FileType    
##  Unknown    :44177   Unknown    :23189   Unknown:23663   java   :13623  
##  Vendor-002: 5314   Country 5 : 9624   G Rank :20098   cs     :10137  
##  Vendor-004: 1121   Country 17: 6513   D Rank : 6284   sql    : 8301  
##  Vendor-003:  872   Country 16: 5330   C Rank : 3051   xml    : 7313  
##  Vendor-006:  594   Country 9 : 3915   B Rank :  333   js     : 6386  
##  Vendor-042:  295   Country 4 : 2008   E Rank :  234   html   : 3393  
##  (Other)   : 1453   (Other)   : 3247   (Other):  163   (Other): 4673  
##          Date          Actual.CE          Developer    
##  06/09/2017: 1267   Min.   :  0.000   81062  :  133  
##  05/09/2017: 1238   1st Qu.:  0.980   79427  :  126  
##  08/08/2017: 1230   Median :  2.610   135746 :  117  
##  18/07/2017: 1219   Mean   :  6.142   75953  :  117  
##  03/08/2017: 1218   3rd Qu.:  6.500   75821  :  109  
##  10/08/2017: 1218   Max.   :943.240   78921  :  107  
##  (Other)   :46436                     (Other):53117  
```

```r
#Create a normalization function
# we can normalize the data

normalize <- function(x) {
return((x - min(x))/(max(x) - min(x)))} # create normalize function
```

```r
ft_new <- aggregate(Date ~ InfraId + FileType, data = ft, length) # taking infraid and Filetype and aggregating it under dates for all developers

ft_new$FileType <- as.numeric(factor(ft_new$FileType , levels = c("cc", "cs", "html", "java", "js", "py", "r", "sh", "sql", "xml")))  # change factor to numeric
```

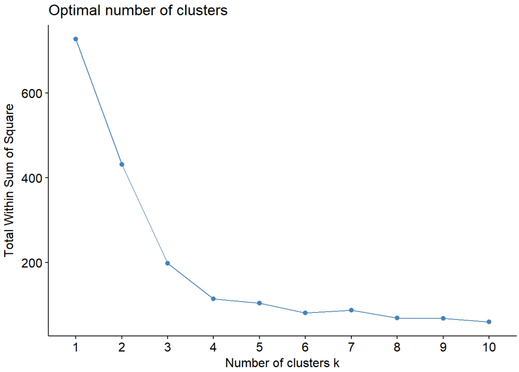
```r
ggplot(ft_new, aes(Date, log(InfraId) , color =  FileType)) + geom_point()
```



```r
# we can observe four classifications for this data
```

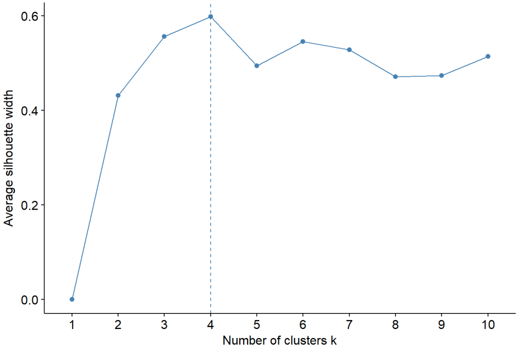
```r
# normalize
norm_ft_new <- as.data.frame(sapply(ft_new, normalize))
```

```r
# Let's check value for the number of clusters to be taken

# Elbow Method
set.seed(1234)

#library("factoextra")

fviz_nbclust(norm_ft_new, kmeans, method = "wss")
```



```r
# silhouette Method
fviz_nbclust(norm_ft_new, kmeans, method = "silhouette")
```

Optimal number of clusters



```r
# We see both methods gives 4 clusters
```

# kmeans

```r
# kmeans
cluster_ft <- kmeans(norm_ft_new, centers = 4, nstart = 25) # with 4 clusters

print(cluster_ft) # to extract the results using 4 clusters
```

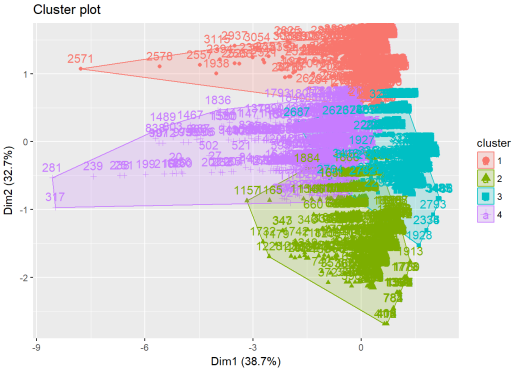
```
## K-means clustering with 4 clusters of sizes 1033, 669, 542, 1244
## 
## Cluster means:
##       InfraId  FileType       Date
## 1 0.06246532 0.9064214 0.03351199
## 2 0.67618134 0.3177213 0.03428650
## 3 0.66876444 0.9081591 0.02073521
## 4 0.06033588 0.2934084 0.05435957
## 
## Clustering vector:
##    [1] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##   [35] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##   [69] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [103] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [137] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [171] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [205] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [239] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [273] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [307] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [341] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [375] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [409] 2 2 2 2 2 2 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [443] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [477] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [511] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [545] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [579] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [613] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 2 2 2 2 2 2 2 2 2 2
##  [647] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [681] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [715] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [749] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [783] 2 2 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [817] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [851] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [885] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [919] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [953] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
##  [987] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1021] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1055] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1089] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1123] 4 4 4 4 4 4 4 4 4 4 4 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1157] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1191] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1225] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1259] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1293] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1327] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1361] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1395] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
```

```
## [1429] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1463] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1497] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1531] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1565] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1599] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 2 2 2 2 2 2
## [1633] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1667] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1701] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1735] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1769] 2 2 2 2 2 2 2 2 2 2 2 2 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1803] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1837] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1871] 4 4 4 4 4 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1905] 2 2 2 2 2 2 2 2 2 1 1 1 1 1 1 1 1 1 1 1 1 1 3 3 1 1 1 1 1 1 1 1 1 1
## [1939] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [1973] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2007] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2041] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2075] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2109] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2143] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2177] 1 1 1 1 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2211] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2245] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2279] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2313] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 1 1 1 1 1 1 1 1 1 1
## [2347] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2381] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2415] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2449] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2483] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2517] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2551] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2585] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2619] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2653] 1 1 1 1 1 1 1 1 1 1 1 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2687] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2721] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2755] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2789] 3 3 3 3 3 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2823] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2857] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2891] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2925] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2959] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2993] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [3027] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [3061] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [3095] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [3129] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [3163] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [3197] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [3231] 1 1 1 1 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3265] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3299] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3333] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3367] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3401] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3435] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3469] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## 
## Within cluster sum of squares by cluster:
## [1] 21.23126 28.52125 18.20710 46.26834
##  (between_SS / total_SS =  84.3 %)
## 
## Available components:
## 
## [1] "cluster"      "centers"      "totss"        "withinss"    
## [5] "tot.withinss" "betweenss"    "size"         "iter"        
## [9] "ifault"
```

```
str(cluster_ft)
```

```
## List of 9
##  $ cluster     : int [1:3488] 4 4 4 4 4 4 4 4 4 4 ...
##  $ centers     : num [1:4, 1:3] 0.0625 0.6762 0.6688 0.0603 0.9064 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:4] "1" "2" "3" "4"
##   .. ..$ : chr [1:3] "InfraId" "FileType" "Date"
##  $ totss       : num 727
##  $ withinss    : num [1:4] 21.2 28.5 18.2 46.3
##  $ tot.withinss: num 114
##  $ betweenss   : num 613
##  $ size        : int [1:4] 1033 669 542 1244
##  $ iter        : int 2
##  $ ifault      : int 0
##  - attr(*, "class")= chr "kmeans"
```

```
# it outputs a cluster of size 1033, 669, 542 and 1244

fviz_cluster(cluster_ft, data = norm_ft_new)  # to visualize the results
```



```
# Extract the clusters and add to our initial data to do some descriptive statistics at the cluster level:

ft_new %>%
  mutate(Cluster = cluster_ft$cluster) %>%
  group_by(Cluster) %>%
  summarise_all("mean")
```

```
## # A tibble: 4 x 4
##   Cluster  InfraId FileType      Date
##     <int>    <dbl>    <dbl>     <dbl>
## 1       1 44605.98 9.157793 13.365924
## 2       2 79238.59 3.859492 13.651719
## 3       3 78820.05 9.173432  8.651292
## 4       4 44485.81 3.640675 21.058682
```